UNIVERSITY OF HELSINKI

AI-assisted physics

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AI-assisted physics with frontier (Oct 2026) models

What current models are good at

Data analysis and software engineering
- Complex data analysis with Python
- Code debugging
- Web (front-end) development
- Shell scripting, job automation

Expert-level mathematicians
- SymPy-verified symbolic calculations
- Numerical verifications

Broad (interdisciplinary) domain knowledge
- Connections to other fields
- Exhaustive literature reviews

Great teachers
- All textbooks squeezed into one reference
- Visualizations, movies, animations, webpages, slide decks

What current models are not good at

Software architecture
- Large and complex software cannot fit in context
- Models are additive, they add code, rather than refactor
- Not enough training data for HPC examples

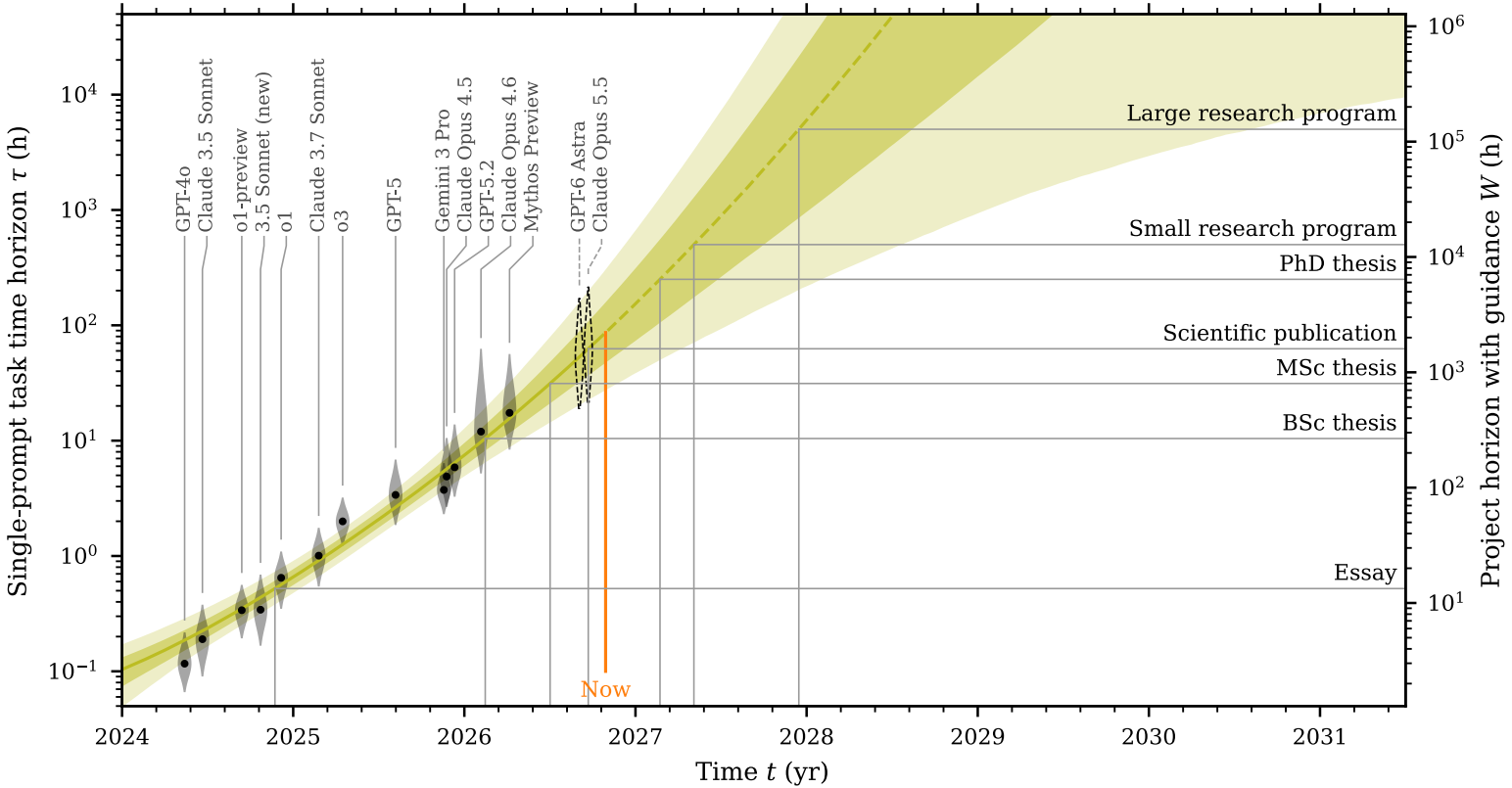
Reliable application of physics
- Hidden assumptions and approximations

Innovation
- Regression to the mean in idea space
- AI does not propose new questions
- No research taste, every paper is treated equally

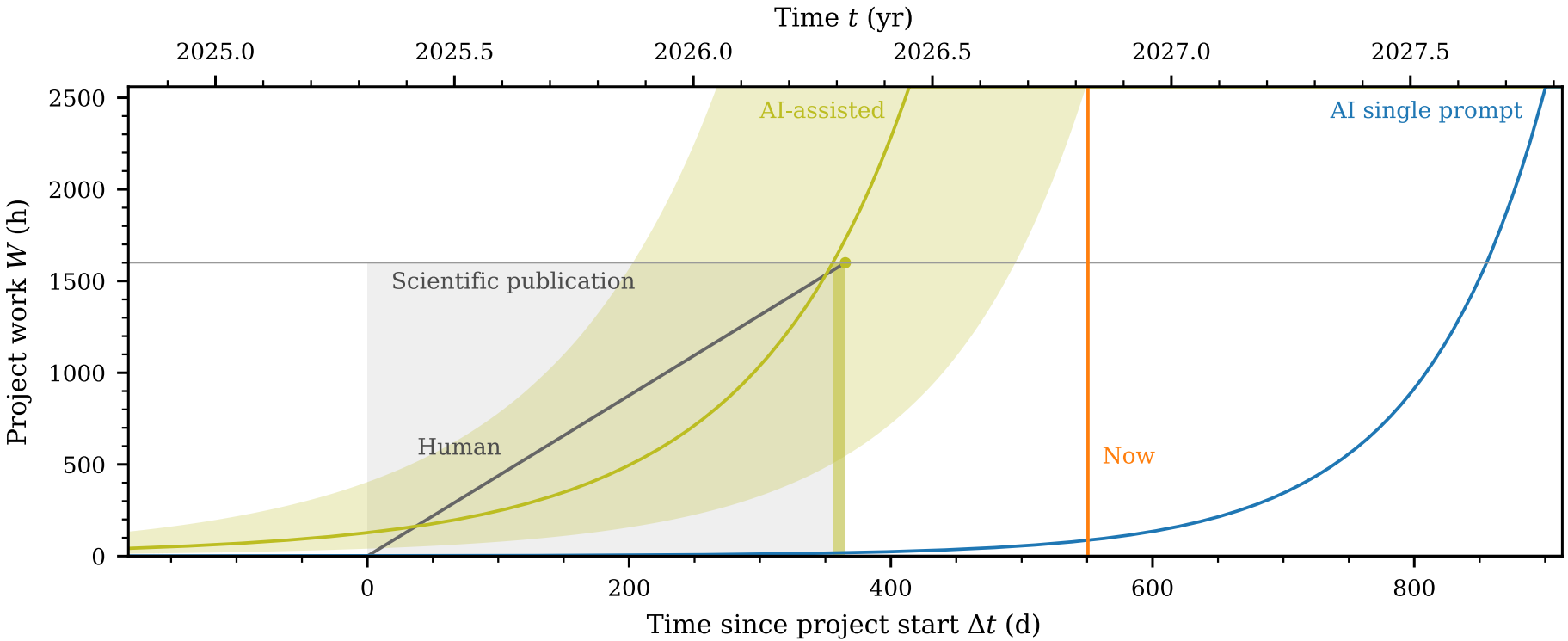
AI capability in physics

Empirical evidence suggests project horizon $W \approx [30 - 300]\tau$ with domain expert guidance

Model capabilities are increasing rapidly.

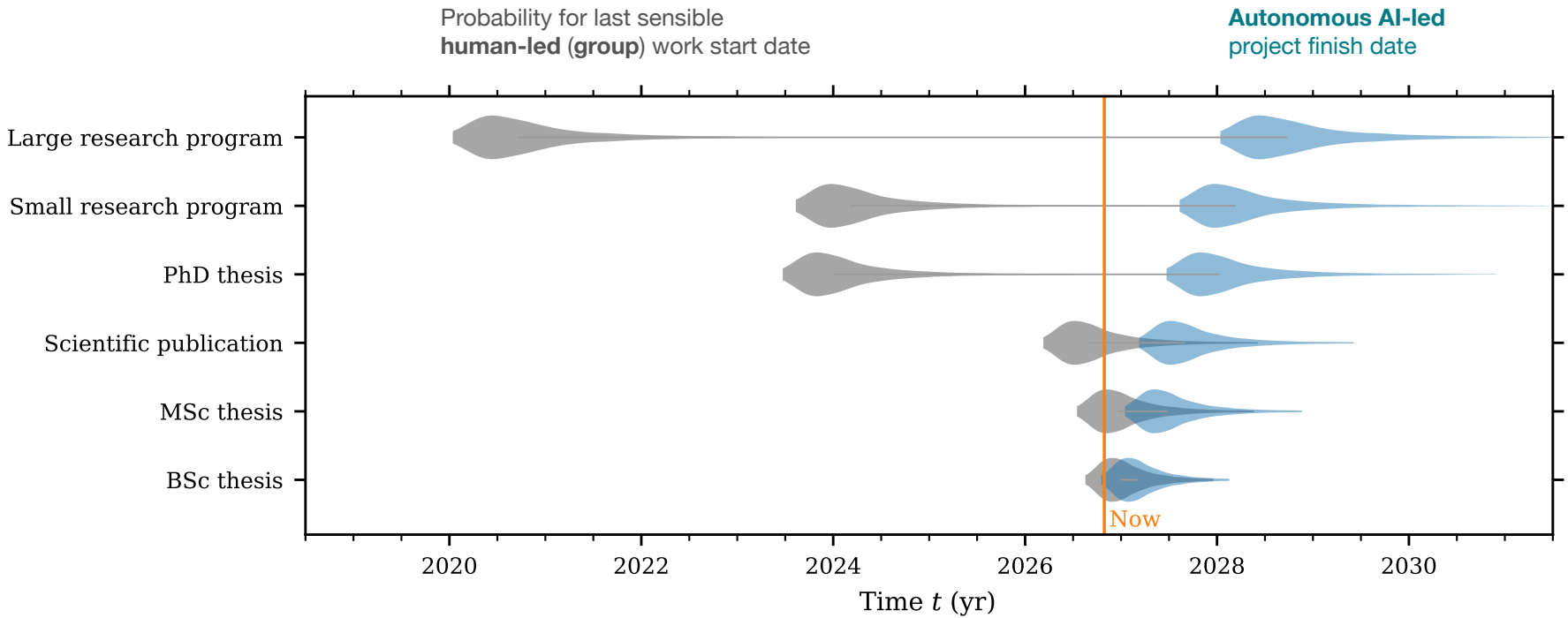


Project execution and the waiting-time paradox



Research horizon

AI-assisted research will soon be the fastest for any research project
(from papers to PhD thesis to multi-million research programs)



New kind of physics

New research environment/infrastructure:

Execution becomes cheap
- Any derivation, simulation or analysis can be tried
- Cost of exploring an idea ≈ zero

Choosing ideas becomes the bottleneck
- Taste is the new scarce resource
- High-risk ideas become rational

Theory–experiment gap widens
- Theory is cheap, data stays scarce
- Unique data and instruments become resources

But, we have to think about:

Verification & validation
- Checking is slower than generating; human verification becomes the bottleneck

Synchronization of results
- What should be published? Failed attempts?
- Papers lose meaning as a unit of productivity
- Can we absorb the flood of incremental results?

Student training
- Expertise is needed to steer and verify, but the execution that built it is automated



i) Technique-based specialization (my code, my method) loses its edge
ii) Data, experiments, infrastructure, and compute become the scarce resource
iii) From execution-limited to **idea-limited physics**